Q A L A

Universal Solution Management OS

UI/UX Specification · Design System · User Flows · Wireframes

Field	Detail
Document Type	Combined UI/UX Specification, Design System, User Flows & Annotated Wireframes
Version	v1.0 — Comprehensive Edition
System	Qala Universal Solution Management OS
Status	Design Review Draft
Personas Covered	14 persona types across all solution domains
Screens Specified	35 primary screens + 22 modal/drawer patterns
Wireframes	28 annotated ASCII wireframes covering all critical screens
User Flows	12 end-to-end flows (onboarding → governance → cross-domain)
Design Principle	Universal architecture · Domain-adaptive vocabulary · Zero software assumptions for non-tech domains
Companion Docs	SDD v2.0 | PRD v2.0 | RFC v2.0 | Requirements v2.0 | Workflows & Use Cases v2.0
Audience	UX Design, Frontend Engineering, Product, Domain Experts, Accessibility Reviewers

1 Design Philosophy

Qala serves pharmaceutical chemists, software engineers, portfolio managers, farm operations managers, tax technologists, and legal operations directors — all on one platform. The UI must feel native to each while being one unified system.

1.1 The Six Design Principles

P1 Universal First, Domain-Native by Persona
Every core concept (solution, version, environment, change, release) is universal. Vocabulary, iconography, and workflow shortcuts adapt per persona. A CPG chemist sees "Batch Release Pipeline" — a developer sees "CI/CD Pipeline" — identical underlying system, domain-appropriate surface.

P2 Progressive Disclosure
The most common action for each persona is one click away. Advanced capabilities (compliance reports, AI model config, cross-factory analytics) are discoverable without being visually dominant. Power users never blocked. New users never overwhelmed.

P3 Trust Through Transparency
Every state change is visible. Every action produces an audit event. The UI makes governance legible — not bureaucratic. Users always know: current state of a solution, who last changed it, what changed, and what happens next.

P4 AI as Companion, Not Gatekeeper
The AI Agent surfaces recommendations inline — dismissable, actionable, contextual. It never blocks a primary workflow. In regulated domains, AI is advisory; human approval is always the final gate. The AI appears when useful, disappears when not.

P5 Accessibility as Architecture
WCAG 2.1 AA is the minimum. Every interaction is keyboard-navigable. Every colour combination meets contrast ratios. Screen reader support is a design constraint, not an afterthought.

P6 Speed as a Feature
The platform manages hundreds of solutions. Data tables load under 200ms. The Command Palette (Cmd+K) is the power-user backbone — available from any screen, returns results in <100ms. Skeleton screens replace spinners for every list and page load.

1.2 Persona-Adaptive Layer

Qala's UI is built on one design system with a persona-adaptive vocabulary layer that activates at login and is adjustable in Profile Settings. It controls three things:

- Vocabulary layer — domain-specific terms replace generic platform terms (see Section 9 for full mapping).
- Widget visibility layer — domain-appropriate dashboard widgets surface for the user's primary domain.
- Quick-action layer — sidebar shortcuts and Command Palette commands pre-populate with domain defaults.

1.3 Responsive Strategy

Breakpoint	Width	Primary Use	Layout
xs	< 640px	Field use — farm, lab, warehouse, delivery	Single column · Bottom tab navigation · Card-first
sm	640–767px	Large mobile, small tablet	Single column · Sidebar as drawer
md	768–1023px	Tablet — lab bench, meetings, client sites	Two column · Collapsible sidebar
lg	1024–1279px	Laptop — developer, analyst, consultant	Full sidebar · Three-column content · All tables
xl	1280–1535px	Desktop — compliance officer, ops centre	Expanded sidebar · Multi-panel · Command palette
2xl	≥ 1536px	Large monitor — analytics, monitoring centre	Ultra-wide multi-panel · Side-by-side comparison

2 Design System — Colour

All colour decisions are token-first. No raw hex values appear in component code. Every colour is referenced by semantic token name, enabling theme switching and accessibility overrides without touching components.

2.1 Brand Palette

	Name	Hex	Usage	WCAG
	Brand Navy	#0D2B45	Primary headings, nav background, logo, brand anchors.	AAA 18.1:1
	Action Blue	#1D6FA4	Interactive elements, links, primary buttons, progress indicators.	AA 5.9:1
	Sky Light	#D6EAF8	Tints, info panels, selected state backgrounds, focus rings.	—
	Sky Mid	#85C1E9	Data visualisation, progress fills, chart elements.	—
	Surface White	#FFFFFF	Primary content surface, card background, dialog background.	—
	Surface Grey	#F4F6F8	Alternate row, secondary surface, sidebar track, divider fill.	—
	Border/Divider	#E2E8F0	Borders, divider lines, disabled component fills.	—
	Text Primary	#1A1A2E	Body text, labels, headings on white surface.	AAA 18.1:1
	Text Muted	#64748B	Secondary labels, captions, metadata, placeholder.	AA 5.2:1

2.2 Semantic Status Palette

	Name	Hex	Usage	WCAG
	Success Dark	#1A5C38	Gate passed, approved, active, deployed states.	AA 10.1:1
	Success Light	#D5F5E3	Success chip background, toast fill, gate-pass panel.	—
	Warning Dark	#7D5A00	Medium risk, deprecated, awaiting review, pending.	AA 7.8:1
	Warning Light	#FEF9E7	Warning chip background, caution panel fill.	—
	Danger Dark	#7B241C	Error state, critical alert, failed gate, recall.	AA 9.1:1
	Danger Light	#FDEDEC	Error chip background, alert panel fill, recall banner.	—
	AI Purple Dark	#4A235A	AI Agent indicator, recommendation badge, intelligence features.	AA 11.3:1
	AI Purple Light	#F5EEF8	AI recommendation card background, insight panel fill.	—
	Teal Dark	#0E6655	Domain extension, use-case headers, feature callouts.	AA 8.7:1

	Teal Light	#D1F2EB	Domain card fill, feature highlight background.	—
	Orange Dark	#B45309	Deprecated state, seasonal event, countdown.	AA 5.0:1
	Orange Light	#FEF3C7	Deprecated chip background, countdown panel.	—

2.3 Semantic Token Map

Token Name	Maps To	Usage Context
color.brand.primary	#0D2B45	Brand identity, global nav, major headers.
color.brand.action	#1D6FA4	All interactive affordances: buttons, links, checkboxes, toggles, focus rings.
color.surface.primary	#FFFFFF	Main content area, cards, dialogs.
color.surface.secondary	#F4F6F8	Alternate rows, sidebar, secondary panels.
color.surface.border	#CBD5E1	All border / divider rules.
color.text.primary	#1A1A2E	Body text, labels, headings.
color.text.secondary	#64748B	Metadata, captions, placeholder, hint text.
color.text.inverse	#FFFFFF	Text on dark surfaces.
color.status.success	#1A5C38	Gate passed, approved, deployed.
color.status.warning	#7D5A00	Medium risk, deprecated, pending.
color.status.danger	#7B241C	Failed, error, recall, quarantined.
color.ai.primary	#4A235A	AI Agent badge, recommendation card.
color.focus.ring	#1D6FA4	3px outline, 2px width, 2px offset — all interactive elements.
color.interaction.hover	#D6EAF8	Row hover, button hover fill.
color.interaction.selected	#D6EAF8	Selected row, active tab, selected chip.

3 Typography & Spacing

3.1 Type Scale

Token	Size	Weight	Line-Height	Usage
display.hero	3.0rem / 48px	800	1.10	Cover, marketing — display only.
display.h1	2.25rem / 36px	700	1.20	Page titles, major section headings.
display.h2	1.75rem / 28px	600	1.30	Section headings, card titles, modal headings.
display.h3	1.375rem / 22px	600	1.35	Subsection headings, panel titles, group labels.
display.h4	1.125rem / 18px	600	1.40	Field group labels, form section headings.
body.large	1.125rem / 18px	400	1.60	Lead body text, onboarding descriptions.
body.base	1.0rem / 16px	400	1.60	Standard body text, form values, table content.
body.small	0.875rem / 14px	400	1.50	Metadata, captions, secondary labels, badges.
body.xs	0.75rem / 12px	400	1.40	Legal micro-text, timestamps, audit footnotes.
label.large	0.875rem / 14px	600	1.40	Button labels, tab labels, form field labels.
label.base	0.75rem / 12px	600	1.30	Badge text, chip labels, status labels.
mono.base	0.875rem / 14px	400	1.50	Code, version numbers, checksums, IDs, hex values.
mono.small	0.75rem / 12px	400	1.40	Inline code, terminal output, digests.

Font family: Inter (via CDN, local fallback: system-ui, -apple-system, sans-serif). Monospace: JetBrains Mono (fallback: ui-monospace, Menlo, Consolas). Font-display: swap. Loading: subsets Latin + Latin-Extended.

3.2 Spacing Scale (8px base grid)

Token	px	rem	Common Usage
space.1	4px	0.25	Icon-to-label gap, chip padding-x tight.
space.2	8px	0.5	Small component internal padding, tight row gap.
space.3	12px	0.75	Badge padding, compact button padding.
space.4	16px	1.0	Standard component padding, card content gap.

space.5	20px	1.25	Form field label-to-input gap.
space.6	24px	1.5	Card padding, modal section gap, sidebar item padding.
space.8	32px	2.0	Section spacing within a page, content group gap.
space.10	40px	2.5	Major section separation, form group gap.
space.12	48px	3.0	Page section breaks, hero area padding.
space.16	64px	4.0	Full-bleed section separation, layout-level gaps.

3.3 Border Radius

Token	Value	Usage
radius.none	0	Tables, dividers, full-bleed elements.
radius.sm	4px	Input fields, small badges, tight chips.
radius.md	8px	Buttons, cards, dropdowns, tooltips.
radius.lg	12px	Modal dialogs, large cards, panel containers.
radius.xl	16px	Hero sections, feature highlight cards.
radius.full	9999px	Pills, status badges, avatar circles, toggles.

3.4 Elevation & Shadow

Level	Definition	Usage
elevation.0	none	Flat — table rows, input fields at rest.
elevation.1	0 1px 3px rgba(13,43,69,0.08)	Default card, sidebar, tooltip anchor.
elevation.2	0 4px 6px rgba(13,43,69,0.07), 0 2px 4px rgba(13,43,69,0.06)	Hovered card, focused input, dropdown origin.
elevation.3	0 10px 15px rgba(13,43,69,0.1)	Dropdown panel, command palette, date picker.
elevation.4	0 20px 25px rgba(13,43,69,0.1), 0 10px 10px rgba(13,43,69,0.04)	Modal dialog, drawer overlay.
elevation.5	0 25px 50px rgba(13,43,69,0.25)	Full-screen overlay, notification centre.

4 Component Library Specification

Every component is defined by: usage context, anatomical breakdown, interaction states, accessibility requirements, and design tokens. Components are built in React with Tailwind token utilities. All components pass WCAG 2.1 AA minimum.

4.1 Navigation Components

4.1.1 GlobalNav — Top Navigation Bar

GlobalNav Layout / Navigation

Usage: Persistent top bar on all authenticated screens. Height: 64px. Contains the primary platform chrome.

Anatomy: Logo (left, 140px) · Search/Command trigger (center, Cmd+K label) · Factory Switcher dropdown · Notification bell (badge count) · AI Agent FAB (purple orb) · User Avatar menu (right).

States: Default · Scrolled (adds elevation.2 shadow) · Mobile collapsed (hamburger → drawer).

Tokens: bg: color.brand.primary · height: 64px · z-index: 100 · logo-width: 140px · avatar-size: 36px

a11y: role="banner", nav has aria-label="Primary", notification bell aria-label="N new notifications", skip-to-content link as first child.

4.1.2 Sidebar Navigation

Sidebar Layout / Navigation

Usage: Left-side contextual navigation. Adapts content based on user persona and active Solution Factory.

Anatomy: Factory selector (top) · Primary nav items (icon + label) · Collapsible sub-groups · Active state indicator bar · Collapse toggle (bottom-left) · Settings + Help links (bottom).

States: Expanded (240px) · Collapsed (64px, icon-only with tooltip) · Mobile (drawer overlay, full-width).

Tokens: bg: color.surface.secondary · active-fill: color.interaction.selected · active-indicator: 3px color.brand.action left border · width-expanded: 240px · width-collapsed: 64px

a11y: role="navigation" aria-label="Main navigation". Active item: aria-current="page". Collapse toggle: aria-expanded. Icons: aria-hidden, labels as accessible names.

Domain-adaptive sidebar items per persona:

Persona	Primary Nav Items	Domain Vocabulary for SDE/Environment
Software Developer	SDEs · Pipelines · Solutions · Tests · Releases · Artefacts	SDE
CPG / Pharma Chemist	Workbenches · Formulations · Batches · QA Records · Releases	Formulation Workbench
Portfolio Manager	Research Workbenches · Funds · Models · Mandates · Compliance	Research Workbench

Farm Ops Manager	Field Environments · Crops · Devices · Yields · Seasons	Field Environment
Principal Consultant	Knowledge Workspaces · Methodologies · Engagements · Catalogue	Knowledge Workspace
Research Scientist	Research Environments · Experiments · Datasets · Publications	Research Environment
Tax Product Manager	Tax Workspaces · Rule Sets · Jurisdictions · Legislative Feed	Tax Workspace
Compliance Officer	Policies · Compliance Reports · CCR Queue · Audit Trail	N/A (no SDE)

4.1.3 Breadcrumb + PageHeader

PageHeader Layout / Context

Usage: Every page shows its hierarchy position. PageHeader adds page-level action buttons.
Anatomy: Breadcrumb trail (up to 4 crumbs, truncated with ellipsis beyond 4) · Page title (H2) · Solution type badge · Lifecycle state chip · Primary action button · Secondary overflow menu.
States: Default · Truncated breadcrumbs · Sticky on scroll (adds shadow) · Mobile (parent crumb only).
Tokens: crumb-font: type.body.small · title-font: type.display.h2 · gap-between-crumbs: space.2 · separator: / in color.text.secondary
a11y: Breadcrumb nav: role="navigation" aria-label="Breadcrumb". Current page: aria-current="page". Overflow button: aria-label="More navigation options".

4.2 Data Display Components

4.2.1 SolutionCard

SolutionCard Data Display / Card

Usage: Primary unit for solution records in grid and list views. Used in dashboards, search results, and factory overview pages.
Anatomy: Solution type icon (24px) · Solution name (title) · Solution type badge · Lifecycle state chip · Version string (mono) · Owner avatar + name · Last-updated timestamp · Kebab overflow menu.
States: Default · Hovered (elevation.2, action bar revealed at bottom) · Selected (color.brand.action border, color.interaction.selected bg) · Loading (skeleton shimmer).
Tokens: bg: color.surface.primary · border-radius: radius.md · padding: space.6 · shadow: elevation.1 → elevation.2 hover · selected-border: 2px color.brand.action
a11y: role="article". Name as heading H3. State chip aria-label="Status: [state]". Avatar aria-label="Owner: [name]". Kebab aria-label="More options for [name]".

4.2.2 LifecycleStateChip

LifecycleStateChip Data Display / Status

Usage: Compact colour-coded indicator of a solution's current lifecycle state. Used inline in cards, tables, headers, and the Transition Panel.

Anatomy: State dot (8px circle, filled) · State label text.

States: Draft (grey #94A3B8) · In Development (blue #1D6FA4) · In Review (amber #7D5A00) · Approved (green #1A5C38) · Released (dark-green #145A32) · Live/Active (teal #0E6655) · Deprecated (orange #B45309) · Recalled (red #7B241C, pulsing dot) · Suspended (red #7B241C) · Retired (dark-grey #475569) · Quarantined (red, dot pulse animation) · Pending Effective Date (amber) · Commissioning (slate #334E68) · Under Review (amber).

Tokens: border-radius: radius.full · padding: space.1 space.3 · dot-size: 8px · font: type.label.base

a11y: aria-label="Status: [state name]". dot: aria-hidden. Colour never sole differentiator — label always visible.

4.2.3 QualityGateProgress

QualityGateProgress Data Display / Progress

Usage: Real-time progress through a release gate checklist. Used on Release detail pages and Release Manager dashboards.

Anatomy: Gate name · Status icon (20px) · Evaluator avatar + name · Timestamp · Evidence artefact link. Vertical connector line between gates.

States: Per gate: Pending (grey circle outline) · Evaluating (blue animated ring) · Passed (green filled checkmark) · Failed (red X) · Waived (orange dash + waiver reason on expand).

Tokens: row-height: 48px min · icon-size: 20px · connector: 2px color.surface.border vertical line · font: type.body.small for metadata

a11y: role="list" aria-label="Release gate checklist". Each gate: role="listitem". Evaluating gates: aria-live="polite" announces state change. Failed gates: aria-describedby links to failure reason.

4.2.4 EventStreamFeed

EventStreamFeed Data Display / Feed

Usage: Real-time log of platform events for an SDE, solution, or factory. The primary activity surface for monitoring and debugging.

Anatomy: Event type icon (16px, colour-coded by category) · Event title · Actor avatar + name · Relative timestamp (absolute on hover) · Expandable event payload panel · Filter bar (event type, actor, time range).

States: Idle (scrollable list) · New event arrives (slide-in from top, 200ms ease-out) · Filtered (empty state with clear-filters CTA) · Empty state.

Tokens: item-height: min 48px · icon-size: 16px · new-event: slide-in 200ms · timestamp-font: type.body.xs · payload: mono.small

a11y: role="log" aria-live="polite" aria-label="Activity feed". New events announced to screen readers. Expandable payload: aria-expanded.

4.2.5 AIRecommendationCard

AIRecommendationCard Data Display / AI

Usage: Displays an AI Agent recommendation inline in dashboards, solution pages, or notifications. Non-intrusive but instantly actionable.

Anatomy: AI orb icon (32px, animated gradient) · Recommendation title · Evidence summary (collapsed by default) · Confidence level bar · Domain context badge · Action buttons: Accept · Dismiss · Defer · Learn More.

States: Default · Expanded (full evidence panel) · Accepted (green confirm animation, card fades) · Dismissed (fade out 300ms) · Loading (orb pulse).

Tokens: bg: color.purpleL · border-left: 4px color.ai.primary · border-radius: radius.md · orb-size: 32px · confidence-bar: 4px height color.ai.primary fill

a11y: role="region" aria-label="AI Recommendation". Dismiss: aria-label="Dismiss recommendation". Accept triggers aria-live announcement. Orb animation respects prefers-reduced-motion.

4.2.6 ArtefactCard

ArtefactCard Data Display / Artefact

Usage: Represents a stored artefact (specification, batch record, attestation, dataset, policy document) in the Artefact Registry.

Anatomy: File type icon · Artefact name · Artefact type badge · Version string · SHA-256 digest (truncated, full on hover) · Created by + timestamp · Download · Verify integrity · View provenance buttons.

States: Default · Verified (green checkmark badge) · Integrity failed (red warning badge + alert) · Loading.

Tokens: bg: color.surface.secondary · digest-font: type.mono.small · badge: border-radius.full · shadow: elevation.1

a11y: Digest: aria-label="SHA-256: [full digest]". Integrity status: role="status". Download: aria-label="Download [name] v[version]".

4.3 Form Components

4.3.1 SolutionTypeSelector

SolutionTypeSelector Form / Selection

Usage: Primary input for choosing a solution type when creating a new solution. Surfaces all 40+ types intuitively across 10 categories.

Anatomy: Category tabs (10 tabs, icons) · Type cards within each category (icon + name + one-line description) · Search filter input (autofocus) · Recently-used strip · Selected type confirmation chip.

States: Default · Category focused · Type selected · Searching · Mobile (full-screen bottom sheet).

Tokens: tab-active: color.brand.action bottom border · type-card: SolutionCard compact variant · search-input: standard TextInput · selected-chip: LifecycleStateChip variant

a11y: Listbox pattern (ARIA listbox + option). Category tabs: tablist / tab. Search: combobox with listbox. Selected chip announces to screen reader.

4.3.2 CCRForm — Change Control Request

CCRForm Form / Multi-step

Usage: Multi-section form for submitting a Change Control Request. Adapts field set based on solution type and change type selected.

Anatomy: Change Type selector · Affected Solutions multi-select (with live cascade impact preview) · Risk Score preview chip (live-computed, colour-coded) · Justification rich-text · Implementation Plan · Rollback Plan · Regulatory Implications section (conditional on solution type) · Approvers panel (auto-computed from routing rules) · Attachments.

States: Filling · Incomplete (inline field-level validation) · Preview mode (read-only review before submit) · Submitted · Approved · Rejected.

Tokens: 2-column layout (lg+), 1-column (md and below) · risk-score-chip: animated colour transition on change · required-indicator: red asterisk + aria-required · attachments: drag-and-drop zone

a11y: role="form". All inputs have associated labels. Error messages via aria-describedby. Risk score updates: aria-live="polite". Required fields: aria-required="true".

4.3.3 DomainMetadataForm

DomainMetadataForm Form / Dynamic

Usage: Dynamic form section rendering domain-specific fields from the solution's Domain Pack JSON schema. No hardcoded fields.

Anatomy: Field definitions loaded from Domain Pack schema at runtime · Supported field types: text, number, date, enum, multi-select, substance lookup (CPG), jurisdiction selector (Legal/Tax), currency (Financial), GPS coordinate (Agricultural), date-range, range-slider.

States: Per-field: Empty · Filled · Invalid (inline error) · Disabled (view-only mode) · Loading (async lookup).

Tokens: Layout: responsive grid auto-fit (minmax 240px, 1fr) · Validation: Zod schema per domain pack · Error: type.body.small color.danger

a11y: All fields have labels. Errors linked via aria-describedby. aria-required. Async lookup: aria-busy. Substance/jurisdiction lookup: combobox pattern.

4.4 Feedback & Status Components

4.4.1 ToastSystem

ToastSystem Feedback / Notification

Usage: Non-blocking notifications for async events: build completed, CCR approved, security alert, AI recommendation available.

Anatomy: Type icon (20px) · Title · Body (optional, 2-line truncated) · Action link (optional) · Dismiss × · Auto-dismiss progress bar.

States: Success (green) · Warning (amber) · Error (red, persistent until dismissed) · Info (blue) · AI (purple). Duration: Success/Info 4s; Warning 6s; Error persistent.

Tokens: position: bottom-right (desktop), top-centre (mobile) · max-width: 400px · border-radius: radius.md · slide-in: 250ms ease-out

a11y: role="status" (Success/Info/AI) or role="alert" (Warning/Error). aria-live="polite" or "assertive" for errors. Dismiss: aria-label="Dismiss notification".

4.4.2 CommandPalette

CommandPalette Navigation / Command

Usage: Power-user command hub accessible from any screen. Returns results in <100ms. Triggered by Cmd+K (Mac) / Ctrl+K (Win/Linux).

Anatomy: Search input (autofocus) · Grouped results: Recent · Solutions · SDEs · CCRs · People · Commands · AI queries (? prefix) · Direct AI commands (! prefix).

States: Open · Searching · Results · No results · Loading.

Tokens: width: 600px · max-height: 600px (scroll) · border-radius: radius.lg · elevation.4 · backdrop: blur(8px) rgba(0,0,0,0.4) · slide-down: 150ms

a11y: role="dialog" aria-label="Command palette" aria-modal="true". Focus trapped. Results: role="listbox". Options: role="option". Esc closes. aria-live result count.

4.4.3 ConfirmationDialog

ConfirmationDialog Feedback / Blocking

Usage: Required for all destructive or high-consequence actions: SDE decommission, solution retirement, recall initiation, emergency change override.

Anatomy: Severity icon (colour-coded) · Title · Explanation paragraph · Consequence bullet list · Confirmation text input (type solution name for critical actions) · Cancel / Confirm buttons.

States: High severity (red accent + confirmation input required) · Medium (amber accent + checkbox required) · Low (standard confirmation button).

Tokens: width: 480px · border-radius: radius.lg · elevation.5 · overlay: rgba(13,43,69,0.5) · input-match-required: red border while no match, green on match

a11y: role="dialog" aria-modal="true" aria-labelledby="dialog-title". Focus trapped. First focus: cancel button (safe default). Confirm disabled until condition met.

5 Wireframes

All wireframes use annotated ASCII notation rendered in monospace. Each wireframe is followed by an element annotation table. Annotations reference the element number shown in the wireframe with [N] labels.

Wireframe Legend
[N] = annotation reference ▇▇▇▇ *= filled element* ▒▒▒▒ *= image / media placeholder*
▓▓▓▓▓▓ *= primary button* ▒▒▒▒▒▒ *= secondary button* ~~~~ *= text content area*
[text] = input field ≡ *= overflow / kebab menu* ● *= status indicator dot*

5.1 WF-01 Login Screen

SC-01 Login Screen | Unauthenticated

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                                                                    |
                                                                    |
                  ╔═══════════════════════════════════╗             |
                  ‖   [1] Q A L A                     ‖             |
                  ‖   Universal Solution Management OS ‖             |
                  ‖ ─────────────────────────────────  ‖             |
                  ‖ [2] ▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓ ‖             |
                  ‖     Continue with your organisation ‖             |
                  ‖ ── or ──────────────────────────── ─‖             |
                  ‖ [3] [ Email address              ]  ‖             |
                  ‖ [4] [ Password              [👁] ]  ‖             |
                  ‖ [5] ☐ Remember me    Forgot pwd? → ‖             |
                  ‖ [6] ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒ ‖             |
                  ‖             Sign in               ‖             |
                  ‖ ─────────────────────────────────  ‖             |
                  ‖ [7] New to Qala? Request access →  ‖             |
                  ╚═══════════════════════════════════╝             |
                                                                    |
```

WF-01 Login Screen — centred card (480px) on Brand Navy full-bleed

#	Element	Specification / Behaviour
1	Logo + Wordmark	Qala logotype + tagline. White on Brand Navy card. Not a link (unauthenticated).
2	SSO Button	Primary button. Full-width. Triggers SAML/OIDC redirect. Pre-fills org domain if detectable. Autofocus on page load.
3	Email Input	TextInput with email type. Autofocus if SSO not detected. aria-label="Email address". Validates on blur.
4	Password Input	TextInput type=password. Eye icon toggles visibility. aria-label="Password". Error shown inline below on invalid credentials.
5	Remember me + Forgot	Checkbox left, link right. Forgot opens email-entry modal (separate flow).
6	Sign In Button	Secondary primary. Disabled until both fields non-empty. Shows spinner + "Signing in..." on submit.
7	New user link	Routes to trial/access request form (public marketing site).

5.2 WF-02 Home Dashboard

SC-03 Home Dashboard | Post-login landing

```
[1] QALA  [Factory: Acme Pharma ▾]  [🔍 Cmd+K]  [🔔 3]  [●AI]  [👤 ▾]

[2] ≡ Home                 [3] Good morning, Elena.  Here's what needs attention:
    Solutions
    Workbenches            [4] ⚠ 3 CCRs          [5] ✗ 1 Gate          [6] ✦ 2 AI
    Batches                  awaiting your         Failure               Insights
    QA Records               approval              Product-X v2.1        ready
    Releases                 [Review Now]          [View Gate]           [View AI]

    AI Insights
    Compliance             [7] Active Formulations                      [View all →]
    Audit Trail
                           ○ Form-A          ○ Form-B          ○ Form-C
    ⚙ Settings           ● In Review       ● Approved        ● Live
    ? Help                 v2.1  Elena       v1.4  Marco       v3.0  Sara

                           [8] Recent Activity                [9] AI Agent Insights
                           ● CCR-042 approved  2m ago         ✦ Form-A stability risk
                           ● Batch B-221 released  1h ago       [Accept] [Dismiss]
                           ● SDE restored  3h ago             ✦ Seasonal model update
                           [Load more →]                        [Accept] [Dismiss]
```

WF-02 Home Dashboard — persona: CPG Formulation Chemist (vocabulary adapted)

#	Element	Specification / Behaviour
1	GlobalNav	Factory name shown in switcher. Bell badge = unread notification count. AI orb animates when recommendations available.
2	Sidebar	Domain-adapted labels. "Workbenches" instead of "SDEs". "Batches" instead of "Artefacts". Active item has 3px left border in Action Blue.
3	Greeting + Priority strip	Greeting uses time-of-day. Priority cards auto-populated from platform state. 3 cards max — surplus handled by notification centre.
4	CCR Approval card	Count badge. Click routes to CCR list pre-filtered to "Awaiting My Approval." Amber accent = action required.
5	Gate Failure card	Solution name + version shown. Click routes to release gate detail. Red accent = blocking.
6	AI Insights card	Count of pending recommendations. Purple accent = AI feature. Click opens AI panel drawer.
7	Active Solutions grid	Last 8 accessed solutions as compact SolutionCards. Domain label = "Active Formulations" for CPG persona.
8	Event Feed	Last 20 events across user's factories. Relative timestamps. "Load more" paginates.
9	AI Recommendation panel	Top 2 recommendations with inline Accept/Dismiss. Accept triggers confirmation then auto-creates CCR.

5.3 WF-03 SDE Detail — Software

SC-05 SDE Detail | Software Development Environment

```
[1] QALA  [Acme Software ▾]  [🔍]  [🔔]  [●AI]  [👤]
```

```
≡ SDEs / my-api-sde
[2] my-api-sde      [3]● Active   v.env-142   [4] Owner: Alex M.   3h ago
[5] [Open in VS Code ▓] [Take Snapshot] [Rebuild] [≡ More]

[6] Overview | Pipeline | Quality | Configuration | Snapshots | Audit Trail

[7] Environment Definition
  Base image: node:20-alpine
  Node: 20.11.0 (pinned)
  pnpm: 8.15.4 (pinned)
  Linter: eslint 8.57 (pinned)
  env-def version: 142 [diff]

[10] Open CCRs on this SDE  (2)
CCR-039  Dep update  ● Approved
CCR-041  Lint config  ● In Review

[8] Live Telemetry
 CPU  ▓▓▓▓▒▒▒▒  42%
 MEM  ▓▓▓▓▓▓▒▒  67%
 Active build: my-api #312  ⟳
 Last build: ✓ PASSED  2h ago

[9] Security Status
 SAST last scan: ✓  4h ago
 CVE check: ✓  No critical
 Baseline policy: ✓ Compliant
```

WF-03 SDE Detail — Overview tab, Software SDE

#	Element	Specification / Behaviour
1	GlobalNav	Standard. Factory name in switcher.
2	SDE Name + Type	SDE name as H2. Type badge hidden on software SDEs (implicit).
3	Lifecycle State Chip	Dot + label. "Active" = teal. Click opens Transition Panel.
4	Owner + Last Modified	Avatar + name. Timestamp relative. Absolute on hover.
5	Primary Actions	"Open in VS Code" is primary for software SDEs. Domain-adapted for other types (e.g. "Open Workbench").
6	Tab bar	6 tabs. Active tab underline in Action Blue. Keyboard: arrow-key nav within tablist.
7	Environment Definition	Tool versions with pinned status. diff link shows changes from previous env-def version.
8	Live Telemetry	CPU/memory bars update every 5s. Active build shows with spinner. Only visible for software SDEs.
9	Security Status	Last SAST/CVE scan result. Green tick = passed. Red X = failed with link to findings.
10	Open CCRs	CCRs linked to this SDE. Clicking a CCR row opens CCR detail in a right drawer.

5.4 WF-04 SDE Pipeline Tab

SC-05 SDE Detail | Pipeline Tab — CI/CD visual

```
Overview | [Pipeline] | Quality | Configuration | Snapshots | Audit Trail

Build #312  ⟳ Running  |  Commit: a3f1c7d  |  Branch: main  |  Started: 3m ago

[1]      [2]       [3]       [4]      [5]       [6]       [7]       [8]
Commit —— Build —— SAST —— SCA —— Test —— Gate —— Attest —— Publish
  ✓        ✓         ✓        ✓        ⟳        ⏸         ⏸         ⏸
```

```
00:08     02:14     01:32     00:44     ↻ 3m…     —         —         —

[9] ▼ Test stage log (live)

 PASS  src/api/routes.test.ts  (14 tests)
 PASS  src/utils/validator.test.ts  (8 tests)
 ↻ RUNNING  src/integration/db.test.ts  ...

[10] ✦ AI suggestion: 3 tests predicted high-risk — defect probability 74%
```

WF-04 SDE Pipeline Tab — live CI/CD view with AI risk overlay

#	Element	Specification / Behaviour
1 – 8	Pipeline Stages	Connected horizontal nodes. ✓ = Passed (green), ↻ = Running (blue animated), ⏸ = Pending (grey), ✗ = Failed (red). Click any node for full log.
9	Stage Log Panel	Live log stream for running stage. Expandable. Copy button. Log filter input.
10	AI Risk Overlay	AI Agent predicts defect probability for test files based on historical change patterns. Dismissable.

5.5 WF-05 Solution Detail Screen

SC-06 Solution Detail | Full lifecycle view

```
≡ Solutions / Product-X / Formulation-A
[1] ⬡ Formulation-A    [2]● In Review    v2.1.0    [3] Owner: Elena V.
[4] [New Version ▓] [Create CCR] [Create Release] [View Registry] [AI ✦] [≡]

[5] Overview | Versions | Changes | Quality | Releases | Distribution | Artefacts

[6] Domain Properties                  [7] Current State
 INCI Name: Aqua                        SDE: Lab-WB-03  [Open]
 Batch type: Liquid emulsion            Spec: formulation-a-v2.1.pdf  [View]
 pH range: 6.0 – 7.5                    Linked CCRs: 3 open
 Regulatory status: EU REACH ✓          Open defects: 1 Medium
 Stability tested: 24 months
 [Edit properties]                      [8] Upcoming Release
                                         Release v2.1.0 → target 12 Mar
                                         Gates: 3/5 passed  ● In Progress
                                         [View Release →]
[9] ✦ AI Insight
Stability prediction: this formulation shows a 14% edge-case risk in
parameter range pH 7.3–7.5 at 40°C. [View Evidence] [Accept] [Dismiss]
```

WF-05 Solution Detail — Overview tab, CPG Formulation solution type

#	Element	Specification / Behaviour
1	Solution Name + Type Icon	Solution type icon (domain-specific). Name as H2, editable inline on click. Type badge.
2	Lifecycle State Chip	Amber "In Review" state. Click opens Transition Panel drawer.
3	Owner + Timestamp	Avatar + name link to user profile. Timestamp shows last modification.
4	Action Bar	Primary action contextual to solution type. "New Version" for Formulation. "New Commit" for software.

5	Tab Bar	6–8 tabs depending on solution type. Domain-specific tabs load additional content (e.g. "Batch Records" for CPG).
6	Domain Properties	Fields rendered from CPG Domain Pack schema. Generic solution detail shows universal fields only.
7	Current State Panel	Right column: active SDE link, current specification artefact, open CCR count, open defect count.
8	Upcoming Release card	Mini release card — version, target date, gate progress. Links to full Release detail.
9	AI Insight Card	AI recommendation specific to this solution. Full evidence on "View Evidence" expand.

5.6 WF-06 CCR Detail Screen

SC-09 CCR Detail | Review & Approval view

```
≡ Change Control / CCR-042  Ingredient Substitution
[1] CCR-042  |  [2] ▇▇ Risk: 8/10 HIGH  |  [3]● Awaiting CCB  |  Elena V.
[4] [Approve ▓] [Request Changes] [Reject] [Escalate to CCB]

[5] Change Description
Substitute Palm Oil Glycerides
with Shea Butter Glycerides in
Products: Form-A, Form-B, Form-C

[7] Implementation Plan
1. Update formulation spec in SDE
2. Run stability simulation x3
3. Upload simulation artefacts
4. Regulatory pre-check

[9] Regulatory Implications
EU REACH: re-notification required
US FDA 21 CFR: no change required

[6] Approval Chain
✓ Stage 1: Formulation Chemist
  Elena V.  Approved  09:14
↻ Stage 2: QA Manager  ← current
  Marco D.  Pending approval
⏸ Stage 3: CCB (4 members)
  Pending stage 2 completion

[8] Impact Analysis
14 affected SKUs
○ Form-A  HIGH impact
○ Form-B  MEDIUM impact
○ Form-C  MEDIUM impact
+11 more...
```

WF-06 CCR Detail — approver view, CPG Ingredient Substitution change type

#	Element	Specification / Behaviour
1	CCR ID + Title	ID is monospace. Title is H2. Breadcrumb shows path.
2	Risk Score	Large chip. 8/10 = red "HIGH". Score breakdown on click/hover. Computed from change type, impact scope, regulatory flag.
3	Lifecycle State	"Awaiting CCB" = amber chip. Click shows available transitions (restrict to authorised roles only).
4	Action bar	Approve, Request Changes, Reject shown only to current stage approver. Other users see view-only actions.
5	Change Description	Rich text (read-only). Implementation plan below.
6	Approval Chain	Vertical timeline. Stage 2 is highlighted as current. Role name + person name per stage.

7	Implementation Plan	Numbered steps from CCR submission. Editable by requester only while in Draft state.
8	Impact Analysis	Solution list with impact severity chips. Mini relationship graph available on "View graph" link.
9	Regulatory Implications	Domain-Pack-driven section. Shows only when solution type has regulatory dimension.

5.7 WF-07 Release Pipeline Screen

SC-10 Release Pipeline | Gate checklist view

```
≡ Releases / REL-0091  Formulation-A v2.1.0
[1] REL-0091  |  Formulation-A v2.1.0  |  [2]● In Progress  |  Target: 12 Mar
[3] [Approve Release ▓] [Download Release Package] [View Artefacts]

[4] Release Gate Progress

 [5]✓ Specification    [6]✓ Regulatory     [7]✓ QA Validation  [8]◌ Stability   [9]⏸ QP Sign-off
 Gate                  Pre-check           Gate                Gate               Gate

 Elena V.  08:20       Compliance AI       QA Mgr  09:41      — Running —        — Pending —
 Artefact ↗            09:12  Pass         Artefact ↗

 ──────────────────────────────── Gate 4: Stability ────────────────────────────
[10] Stability gate is running. Estimated completion: 18 min.

  Stability test suite: 48hr accelerated study (LIMS imported)
  ◌ Test 1: pH stability at 25°C  RUNNING
  ✓ Test 2: Viscosity at 40°C  PASSED
  ✓ Test 3: Microbial challenge  PASSED

[11] AI Release Note Draft:  "Formulation-A v2.1.0: Ingredient substitution..."
 [Edit Draft]  [Approve Draft]
```

WF-07 Release Pipeline — CPG Formulation release, gate 4 running

#	Element	Specification / Behaviour
1	Release ID + Name	Release record ID and target version. Breadcrumb to parent solution.
2	State Chip	● In Progress = amber. Transitions: Approved, Rejected, On Hold.
3	Action bar	"Approve Release" enabled only when all required gates passed and all required sign-offs collected.
4	Gate Progress Strip	Horizontal pipeline. Gates adapt to solution type — CPG gates differ from software gates (no SAST, has Stability + QP).
5 – 9	Gate nodes	✓ = Passed (green). ◌ = Running (blue animated ring). ⏸ = Pending (grey). ✗ = Failed (red). Each shows evaluator and timestamp.
10	Running gate detail	Active gate expands below the pipeline strip showing live test progress.
11	AI Release Notes	AI auto-drafts release notes from linked CCRs and test summaries. Editable. Approval locks the draft.

5.8 WF-08 Formulation Workbench (CPG SDE)

SC-05 Formulation Workbench | CPG SDE type

```
≡ Workbenches / Lab-WB-03  Formulation-A workspace
[1] Lab-WB-03  ● Active  env-def: fd-142  [2] Owner: Elena V.
[3] [Open Simulation ▓] [Import LIMS Results] [Take Snapshot] [≡ More]

Overview | Simulations | Quality Events | Configuration | Snapshots | Audit

[4] Current Formulation
  Version: v2.1.0 (working draft)
  Base: Aqueous emulsion
  Active ingredients: 3
  INCI compliance: ✓ (auto-checked)
  [View full spec]

[5] Quality Events (last 30 days)
  ✓ Stability test  2024-02-10
  ✓ pH verification  2024-02-08
  ⚠ Viscosity out-of-spec  2024-02-05
  ✓ Microbial challenge  2024-02-01

  [Import from LIMS]

[6] Regulatory Auto-checks
  EU REACH:  ✓ Pass
  IFRA:      ✓ Pass
  US FDA OTC: ⏸ Not yet run
  [Run checks now]

[7] Simulation Results
  Sim-031: Substitute simulation
  Result: 11/14 SKUs PASS
  3 SKUs need reformulation
  [View full simulation report]
```

WF-08 Formulation Workbench — CPG persona, domain-adapted SDE

#	Element	Specification / Behaviour
1	Workbench header	Label reads "Workbench" not "SDE". Type adapts to CPG vocabulary throughout.
2	Owner	Avatar + name. CPG workbench shows lab assignment instead of compute quota.
3	Primary action	"Open Simulation" replaces "Open in VS Code". Domain-appropriate primary action.
4	Formulation panel	Current working draft spec. INCI compliance auto-checked on every save.
5	Quality Events	LIMS-imported results shown inline. Out-of-spec results flagged in amber/red.
6	Regulatory Auto-checks	Platform runs regulatory pre-checks on current formulation automatically. Status shown per jurisdiction.
7	Simulation Results	Most recent simulation run with pass/fail summary. Full report opens in a drawer.

5.9 WF-09 Research Workbench — Financial / Quant

SC-05 Research Workbench | Financial SDE type

```
≡ Research Workbenches / RW-Momentum-Q1
[1] RW-Momentum-Q1  ● Active  env-def: rw-088  Owner: Priya S.
[2] [Open Jupyter ▓] [Run Backtest] [Take Snapshot] [Paper Trade] [≡]

Overview | Backtests | Risk Validation | Configuration | Datasets | Audit
```

```
[3] Strategy Environment
  Strategy: Equity Momentum (L/S)
  Universe: MSCI World
  Lookback: 12m
  Rebalance: Monthly
  Training data: v3.2 [checksum ↗]

[4] Performance Dashboard
  Latest Backtest: BT-041  ✓
  Sharpe Ratio:   1.34  ✓ (> 1.0)
  Max Drawdown:   -11.2%  ✓ (< 20%)
  Annual Return:  14.7%
  VaR (99%, 1d):  -2.1%  ✓

[5] Drift Monitor
  Status: ● Stable (last 30d)
  Last alert: 8 days ago

[6] Risk Gate Status
  ✓ Sharpe gate      Pass  [ev ↗]
  ✓ Drawdown gate    Pass  [ev ↗]
  ✓ VaR gate         Pass  [ev ↗]
  ⏸ Liquidity gate   Pending

[7] Regulatory Status
  MAR assessment: ✓ Compliant
  MiFID II algo: ✓ Docs complete
  Pre-trade controls: ✓ Registered
```

WF-09 Research Workbench — Financial / Quant domain, algorithm development

#	Element	Specification / Behaviour
1	Workbench label	Label reads "Research Workbench" — financial domain vocabulary.
2	Primary actions	"Open Jupyter" = primary. "Run Backtest" = secondary. "Paper Trade" = pre-production deployment action.
3	Strategy panel	Model parameters panel. "Training data" shows dataset version with integrity checksum link.
4	Performance Dashboard	Backtest metrics with risk limit compliance indicators (green ✓ / red ✗ per threshold).
5	Drift Monitor	AI drift detection status. Links to drift alert history and recalibration CCR history.
6	Risk Gate Status	Gates specific to financial domain. Each links to evidence artefact.
7	Regulatory Status	MiFID II / MAR compliance indicators. Domain-Pack-driven section.

5.10 WF-10 Agricultural Farm Dashboard (Mobile-first)

SC-13 Agricultural Farm Dashboard | Mobile / Tablet

```
QALA  ≡  |  Acme AgriTech  |  [🔔] [●AI]

[1] FARM MAP
  [F1]▓▓▓▓▓  [F2]▓▓▓▓▓  [F3]▓▓▓▓▓
  [F4]▓▓▓▓▓  [F5]        [F6]▓▓▓▓▓
● F3: Low moisture alert  [View]

[2] IoT Fleet:  150 devices  ●148 online  ○2 offline
Firmware: v2.3 (current)  Last sync: 2m ago

[3] ✦ AI Yield Insight
Season prediction: 4.2 t/ha (target: 4.0)  ✓
⚠ Section F3: irrigation deficit detected
Recommended: increase drip 15% for 14 days
[Accept] [View Evidence] [Defer]

[4] 🏠 Home  🌱 Crops  📡 Devices  📦 Trace  🔍
```

WF-10 Agricultural Farm Dashboard — mobile-first, field operations

#	Element	Specification / Behaviour
1	Farm Map	Field section polygons. Colour-coded by moisture/health. Green = normal, amber = caution, red = alert. Tap section for detail drawer.
2	IoT Fleet status	Device count + online/offline split. Firmware version status. Offline count in red if > 0.
3	AI Yield Insight	Top AI recommendation with Accept/Defer inline. Actionable without leaving dashboard.
4	Bottom Tab Nav	Mobile navigation pattern. 5 tabs. Active tab shown in Action Blue. Tab icons with labels.

5.11 WF-11 Command Palette (Cmd+K)

Command Palette | Overlay on any screen

```
[1] 🔍 [ Search solutions, commands, people...    ]

[2] RECENT
  ⬡ Formulation-A      CPG  ● In Review
  📄 CCR-042           Ingredient Substitution
  🧪 Lab-WB-03         Workbench  ● Active

[3] COMMANDS
  ⊕ New Solution                          Ctrl+N
  📝 Create CCR
  💾 Take SDE Snapshot
  📊 Open Compliance Report

[4] PEOPLE
  👤 Elena Vasquez     Formulation Lead
  👤 Marco Di Lucca    QA Manager

[5] Tip: prefix ? for AI search  ! for AI command
```

WF-11 Command Palette — Cmd+K overlay, default (no search query) state

#	Element	Specification / Behaviour
1	Search Input	Autofocus on open. Instant results (debounced 50ms). Placeholder adapts to persona.
2	Recent group	Last 8 accessed items across solutions, CCRs, workbenches. Arrow-key navigable.
3	Commands group	Platform commands with optional keyboard shortcuts. Domain-specific commands load per persona.
4	People group	Team members with role labels. Click opens user profile or sends notification.
5	AI tip	"?" prefix enables semantic search. "!" prefix sends a command to AI Agent directly.

5.12 WF-12 Compliance Evidence Generation

Compliance Screen | SOC 2 Evidence Package Generation

```
≡ Compliance / SOC 2 Evidence Package
[1] Evidence Package  ● Generating...  |  Scope: All Factories  |  Period: 12m
[2] [Download Package ▒] [Share with Auditor] [Create Regulator Access]

[3] Framework: SOC 2 Type II   Scope: 3 factories   Period: Mar 2024 – Mar 2025

[4] Control Coverage   47 controls total
▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▒▒▒
```

6 User Flows

End-to-end flows for every critical path. Each flow specifies: triggering condition, involved screens, decision points, system responses, events published, and error handling. Flows reference WF wireframes and SC screen IDs.

> **Flow notation: Screen IDs (SC-XX) reference Section 5 specifications. WF-XX references wireframes.**
> *Decision nodes are shown as [D]: indicates the flow branches based on a condition.*
> *Error paths are shown in separate "Error / Alternative" rows below each flow table.*

6.1 UF-01 New User Onboarding — Software Developer

Scope: From invitation email to first successful CI build in a provisioned SDE.

Persona: Software Developer, new to Qala, organisation using SSO.

#	Screen	User Action	System Response	Event / Next State
1	Email (external)	Clicks "Accept Invitation" link.	Browser opens SC-01 Login. SSO button prominent; org domain pre-filled.	SC-01 Login
2	SC-01 Login	Clicks "Continue with [Organisation]."	OIDC redirect to company IdP. On success, JWT issued. Platform detects first-login flag.	SC-02 Wizard Step 1
3	SC-02 Step 1	Selects persona: "Software Developer."	Persona saved to profile. Sidebar and vocabulary layer configured for developer persona.	SC-02 Step 2
4	SC-02 Step 2	Accepts "My Projects" as factory name. Selects "Software Factory" template.	Factory provisioned. Default policies applied (SAST mandatory, attestation required).	SC-02 Step 3
5	SC-02 Step 3	Solution type: Application. Name: "api-gateway". Language: Go.	Solution record created. SDE provisioning starts asynchronously (< 10 min).	SC-02 Step 4
6	SC-02 Step 4	Skips team invitations.	Onboarding marked complete. SDE still provisioning.	SC-03 Dashboard
7	SC-03 Dashboard	Sees "SDE Provisioning" toast with progress bar.	SDE provisions. Toast updates: "api-gateway SDE is ready."	SC-03 (updated)

8	SC-03	Clicks "Open in VS Code" on SDE card.	VS Code opens with Qala Dev extension, SDE auto-connected.	IDE (external)
9	IDE	Makes first code commit to main.	Webhook fires. Pipeline triggered. Event: BUILD_STARTED.	SC-05 WF-04 Pipeline
10	SC-05 Pipeline tab	Watches pipeline — all stages pass.	SLSA attestation created. Quality gate evaluated: PASSED. BUILD_PASSED event.	SC-06 Solution Detail

> *Error Path: SSO fails → inline error + option for email+password (if enabled by org). SDE provisioning fails → toast with error, "Retry" button, support ticket auto-created.*

6.2 UF-02 CPG Ingredient Substitution Change Control

Scope: Supply disruption alert to approved substitution with 14 SKUs updated.

Personas: Supply Chain Director (initiates) · Formulation Chemist (simulates) · QA Manager (validates) · CCB (approves).

#	Screen	User Action	System Response	Event / Next State
1	SC-03 Dashboard	SC Director opens Qala. Navigates to affected formulation via Cmd+K or sidebar.	Platform loads SC-06 Solution Detail for affected product line.	SC-06 Product Line
2	SC-06	Clicks "Create CCR." Selects type: "Ingredient Substitution."	CCRForm opens with Ingredient Substitution field set. Risk preview chip visible.	SC-09 CCR Create (WF-06)
3	SC-09 Create	Fills: original ingredient, substitute, supply disruption rationale, effective date.	Risk score computed live: 8/10 HIGH. Cascade impact query fires in background.	SC-09 (risk updated)
4	SC-09	Reviews impact: 14 SKUs listed with impact severity. Submits CCR.	CCR state → Submitted. CCB + Formulation Chemist notified. Event: CCR_SUBMITTED.	SC-08 CCR List
5	SC-09 (Chemist view)	Chemist opens CCR. Opens Lab Workbench from CCR context link.	Lab Workbench (WF-08) opens in new tab. Substitute properties pre-loaded from CCR context.	SC-05 Workbench (WF-08)
6	SC-05 Workbench	Chemist runs simulation with substitute. 11/14	Simulation artefacts stored. Chemist adds results comment to CCR.	SC-09 CCR Detail (WF-06)

		pass. Uploads artefacts.		
7	SC-09 (CCB view)	CCB reviews CCR + simulation results. All 4 members approve.	CCR state → Approved. Event: CCR_APPROVED. Chemist + SC Director notified.	SC-09 (approved)
8	SC-06 (11 solutions)	Chemist updates 11 formulation specifications. Each update auto-versioned.	11 new spec versions in Artefact Registry, each linked to CCR-042.	SC-06 (×11)
9	SC-06 QA	QA Manager imports LIMS results. Runs validation. Quality gate passed for 11/11.	Quality events ingested. All 11 formulations ready for batch release.	SC-11 Batch Dashboard
10	SC-11	QA Manager triggers batch release for updated formulations.	Batch release workflow initiates (UF-04). Event: BATCH_RELEASE_STARTED.	SC-10 Release (WF-07)

> *Alternative: 3 SKUs requiring reformulation — each triggers a separate CCR with Reformulation change type. Tracked independently on a parallel track.*

6.3 UF-03 Software Blue/Green Production Release

Scope: Tech Lead declares release candidate to live production with rollback capability.

Personas: Tech Lead · Release Manager · QA Engineer · Platform (automated).

#	Screen	User Action	System Response	Event / Next State
1	SC-06 Solution Detail	Tech Lead clicks "Create Release." Selects type: Standard. Links CCRs resolved.	Platform pre-validates: all linked artefacts have valid SLSA attestations.	SC-10 Release (WF-07)
2	SC-10	Platform auto-evaluates gates: SAST, Coverage, No Critical Defects, Attestation.	Gates 1–4 evaluated. All pass. Gate 5 (QA sign-off) pending.	SC-10 (gates 4 passed)
3	SC-10	QA Engineer reviews results. Clicks "Sign off QA approval."	QA approval recorded with timestamp + identity. Compliance evidence record created.	SC-10 (all gates passed)
4	SC-10	Release Manager reviews full release record. Clicks "Approve Release."	Release state → Approved. All approvals collected. Event: RELEASE_APPROVED.	SC-10 (approved)

5	SC-10	Release Manager clicks "Deploy: Blue/Green."	Green environment provisioned. New version deployed to Green. Smoke tests run.	SC-10 (deploying)
6	SC-10	Health metrics validated on Green. Release Manager clicks "Switch Traffic."	Load balancer switches 100% to Green. Blue retained for rollback (48h window).	SC-10 (live)
7	SC-10	Monitors 30-min observation window on SC-10 dashboard.	AI Agent active: monitors error rate, latency p99, business metrics.	SC-10 (monitoring)
8	SC-10	All metrics within threshold. Release Manager confirms stable.	Blue environment scheduled for archive. Release state → Live. Event: DEPLOYED.	SC-06 (Live state)

> *Rollback: If AI Agent detects error rate breach, auto-rollback triggered — traffic returns to Blue within 60 seconds. Release Manager notified. Rollback CCR auto-created. Shown as red alert banner on SC-10.*

6.4 UF-04 CPG Batch Release — QA to Distribution

Scope: Completed production batch through QC testing to retail distribution.

Personas: QA Manager · Release Manager · Distribution Manager.

#	Screen	User Action	System Response	Event / Next State
1	SC-11 Batch Dashboard	Manufacturing submits batch via ERP integration. Batch appears in "Active Batches" tab.	Batch record imported as artefact. Linked to formulation spec version.	SC-11 (active batches)
2	SC-11	QA Manager clicks batch. Opens batch detail drawer.	Batch detail shows: production date, equipment IDs, in-process QC results.	SC-11 (batch drawer)
3	SC-11	QA imports CoA and finished product test results from LIMS.	Quality events ingested. Platform evaluates quality gate: all parameters in spec.	SC-11 (gate evaluated)
4	SC-11	[D] Quality gate passes. QA Manager clicks "QP Sign-off."	QP approval recorded (electronic signature + timestamp). Compliance evidence created.	SC-10 Release (WF-07)
5	SC-10	Release Manager creates batch release record. Links batch ID + spec version.	Release gate check: QP sign-off ✓, spec linkage ✓, distribution plan ready.	SC-10 (all gates)

6	SC-10	Release Manager approves release.	Batch release state → Approved.	SC-10 (approved)
7	SC-10	Distribution Manager initiates distribution: assigns batch to DCs, carrier details.	Distribution record created. GS1 EPCIS events generated. Event: BATCH_DISTRIBUTED.	SC-11 (Released tab)

[D] Gate fail path: Any parameter out-of-spec → QA creates defect. Batch state → On Hold. Investigation workflow initiated. Batch cannot be released until defect resolved and quality gate re-evaluated.

6.5 UF-05 Security Incident — SDE Quarantine and Recovery

Scope: Threat detected to SDE quarantined to clean SDE restored. Full audit trail.

Personas: SEM (automated) · Security Engineer · Developer.

#	Screen	User Action	System Response	Event / Next State
1	Platform (SEM)	SEM detects anomalous egress in developer SDE. Threat score breaches quarantine threshold.	SDE quarantined within 90s. Egress severed. Forensic snapshot captured. Event: SDE_QUARANTINED.	Push notification
2	SC-03 Dashboard	Developer sees red alert banner: "Your SDE api-gateway-sde has been quarantined."	Security Engineer receives push + email notification simultaneously.	SC-05 (quarantined state)
3	SC-05 (quarantined)	Security Engineer opens SDE detail. State chip: QUARANTINED (red, pulsing dot).	Forensic evidence panel displayed. Snapshot accessible in secure viewer.	SC-05 (forensic panel)
4	SC-05 Forensic	Security Engineer reviews process tree, network log, file access log in forensic viewer.	Investigation actions logged in audit trail. Threat vector identified.	SC-05 (investigation)
5	SC-05	Security Engineer clicks "Restore Clean SDE." Selects pre-quarantine snapshot.	New clean SDE provisioned from snapshot. Security baseline scan runs automatically.	SC-05 (provisioning)
6	SC-05	Security baseline passes. Security Engineer confirms restore.	Developer access restored. Developer notified: "SDE restored. New credentials below."	SC-05 (Active, clean)

7	SC-08 CCR	Security Engineer raises emergency CCR to patch compromised package platform-wide.	CVE response workflow triggered across all 47 consumer solutions (UC-TOOL-004 path).	SC-08 (emergency CCR)

6.6 UF-06 AI Recommendation — Accept and Execute

Scope: AI surfaces an environment risk to user accepts to platform auto-creates and executes CCR.
Persona: Research Scientist (Research Environment context).

#	Screen	User Action	System Response	Event / Next State
1	SC-03 Dashboard	AI Recommendation card appears: "Experiment EXP-042 — 3 unpinned dependencies."	AI orb animates. Card shows: title, confidence 87%, evidence link, Accept / Dismiss buttons.	SC-03 (AI card)
2	AI card on SC-03	Scientist clicks "View Evidence."	Evidence panel expands: 3 package names, historical failure rate for similar configs.	SC-03 (card expanded)
3	AI card	Scientist clicks "Accept — Auto-create CCR."	CCR pre-populated: type=Config Fix, affected SDE, recommended change, risk=Low.	SC-09 CCR (prefilled)
4	SC-09 CCR	Scientist reviews pre-filled CCR. Adjusts one version pin. Submits.	CCR submitted. Risk Low = auto-approved. No manual approval required.	SC-09 (auto-approved)
5	SC-05 SDE Detail	Configuration update applied. Environment definition version bumped.	New SDE snapshot. Configuration change event published. AI feedback recorded.	SC-05 (updated)
6	SC-03 Dashboard	AI recommendation card removed. Dashboard returns to normal state.	Feedback stored: accepted + outcome. AI model reinforced positively.	SC-03 (card resolved)

6.7 UF-07 Tax Rule Set — Legislative Update to Effective-Date Release

Scope: New tax legislation to auto-activated rule set on legislative effective date.
Personas: Tax Product Manager · Tax Technologist · Legal · Compliance Officer.

#	Screen	User Action	System Response	Event / Next State

1	SC-03 Dashboard	Tax PM sees AI alert or creates CCR manually: "UK Finance Act 2025 — effective 6 Apr."	CCR created: type=Legislative Update. Routes to Legal + Compliance (parallel review).	SC-09 CCR
2	SC-09 Create	Tax PM fills: 15 jurisdictions, effective dates per jurisdiction, change summary.	15 sub-CCRs auto-generated from master CCR. Each routed independently.	SC-08 CCR List (×15)
3	SC-05 Tax Workspace	Tax Technologist updates rule sets for all 15 jurisdictions in Tax Workspace.	New rule set versions created per jurisdiction. Validation suite runs automatically.	SC-05 (validation running)
4	SC-09 (Legal)	Legal reviews 5 complex jurisdictions. Approves all.	Legal approval recorded per jurisdiction. 10 simpler jurisdictions approved by Tax PM.	SC-09 (approved all)
5	SC-10 Release	Tax PM creates 15 releases. Each with effective-date activation timestamp.	15 Release records. All gates passed. State: Approved, Pending Effective Date. Countdown visible.	SC-10 (×15, countdown)
6	Platform (automated)	At each jurisdiction effective date: release auto-activates.	Rule set deployed per jurisdiction at local midnight. Consumers notified. Event: DEPLOYED.	SC-10 (×15, activated)
7	SC-10	Tax PM reviews: all 15 shown as ● Live.	Compliance evidence of on-time activation stored per jurisdiction. Audit trail complete.	SC-10 (all live)

6.8 UF-08 Fund Mandate Amendment — Investment Governance

Scope: Portfolio Manager proposes EM expansion to IC-approved mandate with investor notification.
Personas: Portfolio Manager · Risk Manager · Compliance Officer · Investment Committee.

#	Screen	User Action	System Response	Event / Next State
1	SC-03 Dashboard	PM sees AI recommendation: "Fund XY — mandate review overdue (180 days)."	PM clicks recommendation. Opens Fund solution record (SC-06).	SC-06 Solution Detail
2	SC-06	PM clicks "Create CCR." Type: Mandate Amendment.	CCRForm opens with Mandate Amendment field set. Risk: 7/10 High. Routes to Risk + Compliance + IC.	SC-09 CCR Create
3	SC-09	PM fills: change rationale, revised	CCR submitted. Risk: 7/10 HIGH. Three-stage	SC-08 (CCR submitted)

		mandate text, investor impact assessment.	approval chain: Risk → Compliance → IC.	
4	SC-05 Research WB (WF-09)	Risk Manager opens CCR. Opens Research Workbench. Runs EM stress test backtest.	Stress test results stored as artefact. Risk gate: pass.	SC-09 (stage 1 done)
5	SC-09 (Compliance)	Compliance Officer: flags 31-day investor notice requirement. Approves.	Compliance approval + notice obligation recorded. Routes to IC.	SC-09 (stage 2 done)
6	SC-09 (IC)	IC reviews at scheduled meeting. 5/5 vote approve.	CCR fully approved. Investor notification pipeline triggered. 31-day countdown starts.	SC-06 (countdown)
7	SC-06	At 31-day notice end: PM activates mandate. Version 2.0 locked.	Mandate v2.0 active. Old mandate archived but queryable. Release event published.	SC-06 (v2.0 active)

6.9 UF-09 Farm-to-Consumer Traceability Query

Scope: Retailer food safety alert to complete provenance report in < 10 minutes.

Personas: Retailer Compliance Team · Platform (automated) · Farm Ops Manager.

#	Screen	User Action	System Response	Event / Next State
1	Traceability Query UI	Retailer opens Qala traceability interface. Enters lot code: LOT-2024-08-221.	Platform initiates provenance query across distribution + batch records.	Loading (< 10 seconds)
2	Traceability Result	Complete provenance chain displayed: consumer lot → 3 input lots → 2 farms → 4 fields.	Interactive provenance tree rendered. Field-level agronomic records accessible.	Traceability Result
3	Traceability Result	Retailer clicks Farm 2, Field 4 node. Reviews harvest date + treatment history.	Field detail retrieved. Cold chain events shown if applicable.	Expanded node detail
4	Traceability Result	Retailer clicks "Export PDF."	Signed PDF generated with chain-of-custody. Export event logged in audit trail.	Export complete
5	SC-13 Farm Dashboard	Farm Ops Manager receives notification: "Traceability query executed by Retailer X."	Notification in event feed. Farm Ops sees what was queried and by whom (transparent audit).	SC-13 (notification)

6.10 UF-10 New Organisation Onboarding — Multi-Domain Setup

Scope: Contract signed to first solutions registered across three domain factories.

Personas: Platform Admin · CTO · Domain Leads (Software, CPG, Service).

#	Screen	User Action	System Response	Event / Next State
1	Admin Portal	Platform Admin provisions new tenant. Sets: isolation tier, data residency (EU), SSO config.	Tenant provisioned. Dedicated infrastructure spun up in specified region.	SC-02 Wizard (admin)
2	SC-02 Admin Wizard	Creates 3 Solution Factories: "Acme Software" · "Acme Consumer Goods" · "Acme Advisory."	3 factories created. Domain-appropriate policies applied per factory type.	SC-04 Factory Overview (×3)
3	SC-04	CTO invites domain leads with pre-assigned roles per factory.	Invitation emails sent. Each lead sees persona-adapted UI on first login.	Email → SC-02 (leads)
4	SC-02 (Software Lead)	Software Lead completes onboarding. Provisions first SDE: Node.js microservice.	Software SDE active. First Application solution registered.	SC-05 SDE (software)
5	SC-02 (CPG Lead)	CPG Lead completes onboarding. Creates Lab Workbench. Imports legacy formulations (CSV).	CPG SDE active. Formulations imported as initial artefact versions.	SC-05 Workbench (CPG)
6	SC-02 (Service Lead)	Service Lead creates first Managed Service. Publishes service specification.	Service solution created. Entry published to internal service catalogue.	SC-06 (Service)
7	SC-03 Dashboard	AI Agent activates across all 3 factories. First recommendations appear within 24 hours.	Cross-factory monitoring begins. AI baseline learning starts per domain.	SC-03 (AI recommendations)

6.11 UF-11 Research Open-Source Library Release

Scope: Research team releases a novel analysis library as governed open source.

Personas: Research Scientist · Principal Investigator · Technology Transfer Office · Legal.

#	Screen	User Action	System Response	Event / Next State

1	SC-09 CCR Create	Scientist raises CCR: type=Open Source Release. Attaches: library name, Apache 2.0 licence, dependency list.	CCR routed to TTO + Legal. Risk = Medium.	SC-09 CCR (TTO+Legal)
2	SC-09 (TTO)	TTO reviews IP ownership. Confirms funded-research IP policy permits release.	IP clearance artefact stored. TTO approval recorded.	SC-09 (stage 1 done)
3	SC-09 (Legal)	Legal reviews all dependency licences for Apache 2.0 compatibility.	SBOM generated. All licences verified compatible. Legal approval recorded.	SC-09 (stage 2 done)
4	SC-05 Research Env	Scientist adds LICENCE, CITATION.cff, docs. Commits. Pipeline runs and attests.	SLSA-equivalent attestation generated for source release artefact.	SC-05 (attested)
5	SC-09 (PI)	Principal Investigator gives final approval.	CCR fully approved. Release state → Approved.	SC-10 Release
6	SC-10	Platform distributes to PyPI / CRAN / npm. DOI assigned via DOI service.	Library published. Distribution record created. Event: SOLUTION_RELEASED (open_source).	SC-06 (Open Source state)
7	SC-06	Library now governed: version lifecycle, automated security scans, consumer notifications.	Library managed as Library solution type. Future versions require governed pipeline.	SC-06 (ongoing)

6.12 UF-12 Compliance Examination — Regulator Read-Only Access

Scope: Regulatory examination notice received to auditor completes review via Regulator Access portal.

Personas: Chief Compliance Officer · External Regulator (read-only).

#	Screen	User Action	System Response	Event / Next State
1	SC-03 Dashboard	CCO receives examination notice. Opens Compliance section.	Compliance overview shows current posture across all factories.	Compliance Overview
2	Compliance Screen (WF-12)	CCO selects: framework=SOC 2 Type II, scope=all	Evidence generation queued. Est. time: 6 minutes shown.	Evidence generating

		factories, period=12 months.		
3	Compliance Screen	Evidence package complete. CCO reviews: 43/47 controls evidenced, 4 need supplement.	CCO uploads 4 supplementary artefacts for partial controls.	Evidence complete
4	Compliance Screen	CCO clicks "Create Regulator Access." Sets: 30-day duration, read-only, SOC 2 scope.	Regulator access portal created. Auditor receives email with secure one-time link.	Regulator Access portal
5	Regulator Portal	Auditor logs in. Browses evidence, downloads artefacts, queries audit trail.	All auditor actions logged in immutable audit trail. CCO sees auditor activity in real time.	Auditor read-only view
6	Regulator Portal	Auditor completes review after 5 days. Raises 2 minor observations.	Observations recorded as comments linked to specific controls.	Regulator Portal (completed)
7	Compliance Screen	CCO revokes Regulator Access session. Downloads final evidence package for records.	Session revoked. Full interaction log archived. SOC 2 audit record complete.	Session revoked

7 Interaction Patterns & Micro-interactions

7.1 Lifecycle State Transition Panel

Clicking any LifecycleStateChip opens a right-side transition drawer (never a full modal — transition is consequential but not blocking):

- Current state shown prominently at top.
- Available next states shown as cards — greyed if preconditions not met.
- Precondition blockers shown inline (e.g. "1 critical defect must be resolved before Approved → Released").
- Confirm button activates only when a valid transition is selected.
- Emergency transitions (Quarantine override, Recall initiation) require a typed justification of at least 20 characters before Confirm activates.
- Successful transition: state chip pulses with brief colour animation. AUDIT_EVENT published immediately.

Transition Panel | Right-side drawer on state chip click

```
Solution: Form-A
● In Review
Click chip ──────►

State Transition
────────────────────────
Current: ● In Review
────────────────────────
Available transitions:

[✓ Approve — move to Approved]
Requires: QA sign-off ✓
Requires: All CCRs resolved ✓

[✗ Request Changes — to Draft]
Add reason (required):
[ ──────────────────────── ]

[Recall — blocked]
⚠ Only QA Director may recall

[▓ Confirm transition ▓]
```

Transition Panel — right-side drawer, inline precondition checking

7.2 Optimistic UI Updates

Action	Optimistic Behaviour	Rollback if Server Fails
CCR approval (own stage)	Approval chip shows "Approved" immediately.	Reverts to "Pending" + error toast.
Dismiss AI recommendation	Card fades out immediately (300ms).	Card reappears + error toast.
Solution name inline edit	Name updates immediately on blur.	Reverts to original + inline error.
Snapshot trigger	Snapshot row appears as "In Progress" instantly.	Row removed + error toast.

Pipeline trigger	Pipeline status changes to "Running" immediately.	Reverts + error notification.
Tag assignment	Tag chip appears on card immediately.	Chip removed + error toast.
State transition (low-risk)	State chip updates immediately on confirm.	Chip reverts + error dialog.

7.3 Loading States

Context	Pattern	Trigger
Page load	Skeleton screen — grey shimmer blocks matching content layout.	Always on page navigation.
Data table	Skeleton rows — 5 rows at target height, animated shimmer.	Always on data fetch.
Action button	Button → spinner icon + disabled state + "Processing..." label.	On any async submit.
AI analysis	AI orb animated pulse + "Analysing…" label.	During AI computation.
SDE provisioning	Step progress bar: Initialising → Building → Configuring → Ready.	Always (can take up to 10 min).
Build pipeline stages	Animated stage node border (blue rotating ring).	While stage running.
Evidence generation	Percentage ring + "Collecting evidence for N controls" label.	Always (typically 2–8 min).
Backtest execution	Progress bar with estimated time + stage label (Loading data, Running, etc.).	While backtest running.

7.4 Empty States — Domain-Adaptive

Screen Context	Illustration	Message	CTA
No SDEs (Software)	Blueprint grid illustration	Your factory has no environments yet. Provision an SDE to start.	"Provision SDE"
No SDEs (CPG)	Lab flask illustration	No formulation workbenches yet. Set up your first lab workspace.	"Create Workbench"
No SDEs (Financial)	Chart + magnifier	No research workbenches yet. Create your first quantitative environment.	"Create Workbench"
No search results	Binoculars illustration	No solutions match your filters.	"Clear filters"
No CCRs pending approval	Checkmark + tea cup	Nothing waiting for your approval. Enjoy the quiet.	None — positive state
No AI recommendations	AI orb sleeping	No recommendations right now. The AI Agent is monitoring.	None — reassuring

No events in feed	Clock illustration	No events in this period. Activity will appear here as solutions evolve.	"View all events"
Compliance: no gaps	Shield + checkmark	All controls evidenced. Your compliance posture is strong.	"Download Report"

7.5 Keyboard Navigation Map

Shortcut	Action
Cmd/Ctrl + K	Open Command Palette
Cmd/Ctrl + N	New Solution (context-aware to current factory/type)
Cmd/Ctrl + /	Open AI Assistant chat panel
Cmd/Ctrl + B	Toggle sidebar collapsed / expanded
Cmd/Ctrl + Shift + F	Open global Search page
G then H	Go to Home Dashboard
G then S	Go to Solutions list
G then E	Go to Environments / SDEs / Workbenches
G then C	Go to CCR queue
G then R	Go to Releases
G then A	Go to Audit Trail
G then I	Go to AI Insights panel
?	Open keyboard shortcut reference modal
Esc	Close modal / drawer / palette / inline editor
Enter (on row)	Open detail view for the focused row
Space (on row)	Select / deselect row for batch operations
Arrow keys	Navigate within dropdowns, command palette groups, tab bars
Tab / Shift+Tab	Move between focusable elements (all interactive elements reachable)

8 Accessibility Specification

WCAG 2.1 Level AA minimum. The platform must be fully operable for users with visual, motor, and cognitive differences. This section defines implementation requirements for every major component and interaction pattern.

8.1 Colour & Contrast

Context	Requirement	Implementation
Body text on white	Contrast ≥ 7:1 (AAA target)	Text Primary #1A1A2E on #FFFFFF = 18.1:1 ✓
Secondary text on white	Contrast ≥ 4.5:1 (AA min)	Text Muted #64748B on #FFFFFF = 5.2:1 ✓
Interactive elements	Contrast ≥ 4.5:1	Action Blue #1D6FA4 on #FFFFFF = 5.9:1 ✓
Success text on light bg	Contrast ≥ 4.5:1	Success Dark #1A5C38 on #D5F5E3 = 5.1:1 ✓
Warning text on light bg	Contrast ≥ 4.5:1	Warning Dark #7D5A00 on #FEF9E7 = 4.8:1 ✓
Danger text on light bg	Contrast ≥ 4.5:1	Danger Dark #7B241C on #FDEDEC = 5.4:1 ✓
Focus ring	3px outline, 2px width minimum	#1D6FA4 focus ring — never suppressed by CSS.
Colour not sole indicator	Never use colour alone for state	All status chips: colour + icon + text label. Lifecycle dots: colour + aria-label.

8.2 ARIA & Keyboard Requirements

Requirement	Implementation Pattern
Tab order matches DOM order	No positive tabindex. Logical DOM order equals visual order. Sidebar items in display order.
All interactive elements keyboard-reachable	Every button, link, input, select, and custom control reachable via Tab.
Focus always visible	Never outline:none without custom replacement. Custom :focus-visible ring on all interactive elements.
Skip to content	First focusable element on every page, visible on focus, jumps to main content. aria-label="Skip to main content".
Page landmarks	main, nav, aside, header, footer — all present. Multiple same-type landmarks have unique aria-label.
Heading hierarchy	One H1 per page. Logical hierarchy, never skipping levels (H1→H2→H3, not H1→H3).
Modal / dialog focus trap	All modals trap Tab. Focus moves to modal on open. Returns to trigger on close.
Drawer focus management	Focus moves to first interactive element in drawer on open. Returns to trigger on close.

Live regions	aria-live="polite" for toasts, status updates, AI recommendations. aria-live="assertive" for security alerts only.
Data tables	All data tables: th with scope. Complex tables: aria-describedby to summary. No layout tables.
Form labels	Every input: label or aria-label or aria-labelledby. Errors: aria-describedby links to error message.
Loading states	aria-busy="true" on loading containers. aria-live region announces "Loading complete."
Icon-only buttons	All icon-only buttons: aria-label with descriptive text. SVG icons: aria-hidden="true".
Combobox pattern	All type-ahead selects (SolutionTypeSelector, CCR affected solutions): ARIA combobox + listbox pattern.
Drag and drop	All DnD (attachment uploads, reorder) has keyboard equivalent. No DnD-only interaction.

8.3 Motion & Cognitive Accessibility

- prefers-reduced-motion respected universally. All CSS transitions replaced with instant or opacity-only when set. AI orb animation, pipeline stage animations, slide-in toasts all affected.
- Plain language: All UI copy at Grade 8 reading level or below for transactional text. Technical terms explained in tooltips.
- Descriptive errors: Always explain what went wrong + how to fix it. Never "Error 400." Always "The email you entered is not in a valid format — please check and try again."
- Timeout warning: Sessions expire with a 5-minute modal warning. Extending session does not lose work (all form drafts auto-saved every 30 seconds).
- Consistent navigation: Sidebar structure never changes mid-session. Same position, same order every page.
- No time limits on forms. CCR drafts auto-saved. Completion resumed on next login.

8.4 Mobile & Touch

Requirement	Implementation
Touch target minimum	44×44px for all interactive elements. 8px minimum spacing between targets.
Swipe gesture fallback	No swipe-only actions. All swipe affordances have tap equivalents.
200% zoom support	All screens functional at 200% text zoom with no horizontal overflow.
Both orientations	All screens functional in portrait and landscape.
iOS Dynamic Type	Font sizes respect system text size settings on iOS Safari.
Haptic feedback	Used sparingly for confirmation of critical actions only (iOS: UIImpactFeedbackGenerator.heavy for Quarantine, Recall).

9 Domain Vocabulary Reference

The UI renders domain-appropriate vocabulary based on the user's primary persona and the solution type currently in view. This reference defines the canonical term mapping for every platform concept across all 8 major domains.

Platform Concept	Software / Tech	CPG / Pharma	Financial	Agricultural	Services	Res
Development Environment	SDE / Dev Env	Formulation Workbench	Research Workbench	Field Environment	Knowledge Workspace	Res
Solution	Application / Service	Formulation / Product	Fund / Model	Agri-Solution	Methodology / Service	Res
Build / Compile	Build	Simulation Run	Backtest	Seasonal Analysis	Engagement Prep	Exp
CI/CD Pipeline	CI/CD Pipeline	Batch Release Pipeline	Risk Validation Pipe	Seasonal Pipeline	Delivery Pipeline	Exp Pipe
Artefact	Container Image / Pkg	Specification / CoA	Model Artefact	Protocol / Record	Deliverable / Method	Data Res
Quality Gate	CI Quality Gate	Batch Release Gate	Risk Gate	Agronomic Gate	Delivery Quality Gate	Pee Gate
Release	Software Release	Batch / Product Launch	Fund Activation	Seasonal Release	Service Catalogue Add	Pub
Distribution	Deploy / Publish	Retail Distribution	Fund Distribution	Farm Distribution	Service Delivery	Ope Pub
Version	v1.2.3 (semver)	Spec Version	Model Version	Protocol Version	Methodology Version	Data
Change Control (CCR)	Change Control	Change Control	Model Change	Protocol Update	Methodology Update	Prot
Lifecycle State	Draft→Released →Live	Formulation→Ba tch→Dist	Research→Live →Retired	Pilot→Active→A rchived	Design→Approv ed→Published	Draf d→F
Owner	Developer / Tech Lead	Formulation Chemist	Portfolio Manager	Farm Ops Manager	Principal Consultant	Lead Res

Implementation note: The vocabulary layer is loaded at login from the user's persona profile and reloads when the user switches the active Solution Factory. Custom vocabulary can be defined per organisation via the Localisation admin panel.

10 Additional Wireframes

Supplementary wireframes covering modal patterns, the AI agent panel, the audit trail view, and mobile navigation.

10.1 WF-13 CCR Impact Analysis Panel

CCR Impact Analysis | Right-side panel within CCR Detail

```
Impact Analysis  ×
CCR-042  Ingredient Substitution

[1] Impact Scope
14 affected solutions
Across 3 product lines

[2] Impact by Severity
 HIGH   3 solutions
 MED    6 solutions
 LOW    5 solutions

[3] Affected Solutions
○ Form-A      HIGH  [View →]
○ Form-B      MED   [View →]
○ Form-C      MED   [View →]
○ Form-D      LOW   [View →]
+10 more      [Show all]

[4] [View Relationship Graph]
```

WF-13 Impact Analysis Panel — right drawer within CCR Detail

#	Element	Specification / Behaviour
1	Impact Scope	Total count + product line grouping. Computed in background on CCR submission.
2	Severity bars	Stacked bar breakdown. Click any bar filters solution list below.
3	Solution list	Shows top 4. Each row: solution type icon, name, severity chip, View link. "Show all" expands.
4	Graph view	Opens interactive relationship graph in a new panel — nodes clickable, zoom/pan.

10.2 WF-14 AI Agent Insight Panel

AI Agent Panel | Right-side drawer (●AI button in GlobalNav)

```
✦ AI Agent — Insights    ×
[1] Factory: Acme Pharma   [All solutions ▾]

[2] Active Recommendations  (5)

✦ HIGH CONFIDENCE
Form-A: edge-case pH instability risk
```

```
Confidence: 91%  |  Domain: Quality
Detected: parametric drift in pH 7.3-7.5
[View Evidence] [Create CCR] [Dismiss]

✦ MEDIUM CONFIDENCE
Lab-WB-03: 2 unpinned dependencies
Confidence: 78%  |  Domain: Environment
[Accept Fix] [Dismiss]

[3] Recent AI Actions
✓ CCR-039 auto-created  2h ago
✓ Recommendation dismissed  1d ago

[4] [Open full AI insights dashboard →]
```

WF-14 AI Agent Panel — collapsible right drawer

#	Element	Specification / Behaviour
1	Scope selector	Factory and solution type filter. "All solutions" shows platform-wide. Narrow to a single solution for focused view.
2	Recommendation list	Sorted by confidence descending. Each card: category, confidence bar, brief evidence summary.
3	Recent AI Actions	Audit trail of AI-triggered events. Creates trust through transparency of what AI has done.
4	Full dashboard link	Routes to the AI Insights full-page view with charts, trend analysis, and feedback history.

10.3 WF-15 Audit Trail Screen

Audit Trail | Immutable event log view

```
≡ Audit Trail
[1] Filter: [ Solution name or ID   ] [Event Type ▾] [Actor ▾] [Date Range ▾]
[2] [Export CSV] [Export JSON] [Generate Compliance Report]

[3] 3,847 events   Showing: last 30 days   Filtered: CCR events

[4] 2025-03-07  09:14:22  CCR_APPROVED    CCR-042  Elena Vasquez  → QA stage
    [Expand ▼]

    Event ID: evt_9a3f2c1d  |  Facility: EU-WEST-1  |  IP: [masked]
    CCR-042: Ingredient Substitution  Risk: 8/10
    Approver: Elena Vasquez  Stage: 1 of 3  Approval note: "Simulation pass"

2025-03-07  09:02:15  ARTEFACT_PUBLISHED  Form-A-spec-v2.1  System
2025-03-07  08:57:43  CCR_SUBMITTED       CCR-042  Marco Di Lucca
2025-03-07  08:32:11  SDE_SNAPSHOT        Lab-WB-03  Elena Vasquez
2025-03-07  07:55:29  BUILD_PASSED        api-gateway #312  CI Agent
[5] ← Prev     Page 1 of 193     Next →
```

WF-15 Audit Trail — immutable event log with expand and export

#	Element	Specification / Behaviour
1	Filter bar	Real-time filter. Event type multi-select. Actor autocomplete. Date range picker (calendar).

2	Export actions	CSV and JSON exports honour current filter. Compliance report generates a formatted evidence document from filtered events.
3	Result summary	Total count + active filter description. Helps user understand the scope of visible records.
4	Event row	Timestamp (ISO 8601) · Event type (monospace badge) · Resource ID (link) · Actor · Summary. Expand shows full event payload.
5	Pagination	Page-based for predictable performance on large audit trails. Alt: virtual scroll on lg+ screens.

11 Mobile Patterns & Progressive Web App

11.1 Mobile Navigation

On mobile (xs/sm breakpoints), the sidebar is replaced by a bottom tab bar with 5 primary destinations. Secondary navigation is accessed via the hamburger drawer triggered by the ≡ icon.

Tab	Icon	Label	Tap Destination
1	🏠 Home	Home	Home Dashboard (SC-03)
2	🔵 Solutions	Solutions	Solution list (SC-06 list view)
3	⊕ New	New	Bottom sheet: New Solution / New CCR / New SDE
4	🔔 Activity	Activity	Notification centre + event feed
5	👤 Profile	Profile	User profile + settings + help

11.2 Progressive Web App Configuration

- Service Worker: caches critical assets and recent solution data for offline read-only access (farm field use case).
- Web App Manifest: installable on home screen. Standalone display mode. Domain-appropriate app name and icon.
- Push Notifications: opt-in for CCR approvals, security alerts, build status, AI recommendations. Respects OS notification settings.
- Background Sync: CCR draft form submissions queued when offline, auto-submitted when network reconnects.

11.3 Offline Capabilities

Feature	Online	Offline
View solution records	Full detail + live data	Last cached version + amber "Offline" banner
View field telemetry	Live IoT sensor stream	Last 24h cached data + sync timestamp
Create CCR draft	Live risk score + routing preview	Draft saved locally, submitted on reconnect
Traceability query	Full live provenance query	Not available — requires network (shown clearly)
AI recommendations	Live AI analysis	Cached last-sync recommendations shown
Audit trail	Full live query	Not available — requires network

12 Design Handoff Notes

12.1 Component Implementation Notes

- All components built in React 18 + TypeScript. Styling via Tailwind CSS with custom design token plugin.
- Component library: exported as npm package "@qala/design-system". Storybook hosted internally for interactive component reference.
- Persona vocabulary layer: implemented as a React Context (PersonaContext) wrapping all routes. Components consume vocabulary via usePersonaVocab() hook.
- Domain metadata forms: rendered by DomainFormRenderer component consuming JSON schema from Domain Pack API. No code changes required for new domains.
- Animation: Framer Motion for all transitions. All animations wrapped in a MotionProvider that checks prefers-reduced-motion.

12.2 Token Implementation

Design tokens are defined in a single source of truth (JSON) and compiled to: CSS custom properties, Tailwind theme extension, and React theme object.

```
// CSS custom properties (design-tokens.css)
--color-brand-primary: #0D2B45;
--color-brand-action: #1D6FA4;
--space-4: 1rem;
--radius-md: 0.5rem;

// Tailwind config extension
theme.extend.colors["brand-primary"] = "var(--color-brand-primary)"
theme.extend.spacing["4"] = "var(--space-4)"
```

12.3 Persona Vocabulary Configuration

Each persona vocabulary set is defined as a JSON file loaded from the platform configuration API:

```
// PersonaVocab type (simplified)
{
  "sde":          "SDE",               // Software persona
  "sde":          "Formulation Workbench", // CPG persona
  "build":        "Build",
  "build":        "Simulation Run",    // CPG persona
  "release":      "Software Release",
  "release":      "Batch Release",     // CPG persona
}
```

12.4 Figma File Structure

Figma Page	Contents
01 — Foundations	Colour tokens, typography scale, spacing grid, elevation, radius, iconography.
02 — Components	All components from Section 4. One frame per component. All states documented.
03 — Wireframes	All annotated wireframes from Section 5. Low-fidelity greyscale.
04 — Screen Library	All 35 primary screens in high-fidelity per persona (Software, CPG, Financial, Agri, Research).
05 — User Flows	All 12 user flows as connected screen sequences with decision nodes.
06 — Mobile	All screens at 390×844px (iPhone 14). Breakpoint variants for md/lg also documented.
07 — Dark Mode (TBD)	Planned — token inversion. Not in scope for v1.0.

End of Document — Qala UI/UX Specification & Design System v1.0